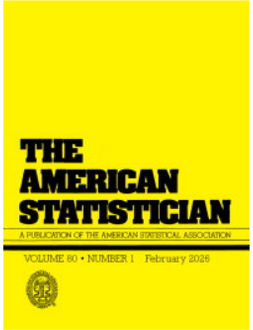

The American Statistician

ISSN: 0003-1305 (Print) 1537-2731 (Online) Journal homepage: www.tandfonline.com/journals/utas20

Probabilistic Parameter Estimates that Require Less Small Print

James A. Hanley

To cite this article: James A. Hanley (06 Mar 2026): Probabilistic Parameter Estimates that Require Less Small Print, The American Statistician, DOI: 10.1080/00031305.2025.2606079

To link to this article: https://doi.org/10.1080/00031305.2025.2606079

View supplementary material

Published online: 06 Mar 2026.

Submit your article to this journal

Article views: 99

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=utas20

THE AMERICAN STATISTICIAN
2026, VOL. 00, NO. 0, 1–7: Teacher's Corner
https://doi.org/10.1080/00031305.2025.2606079

Taylor & Francis
Taylor & Francis Group

Check for updates

Probabilistic Parameter Estimates that Require Less Small Print

James A. Hanley

Department of Epidemiology, Biostatistics, and Occupational Health, McGill University, Montréal, Québec, Canada

ABSTRACT
Although we have had nearly a century to refine it, our teaching of confidence intervals for parameters is still imperfect. Despite all of our warnings regarding these intervals, it is not uncommon for end-users to misinterpret them. We discuss some possible reasons for this, and using a printed figure and a `Shiny` app, work through a simple and close-to-home example while trying to avoid many of these traps. We urge teachers to (a) begin with contexts that require less technical knowledge, or where the technical details can be kept out of the way (b) avoid the traditional (and symmetric) "point estimate ± a *z*- or *t*-based margin of error" confidence intervals that lead to lazy and muddled thinking (c) start with a direct approach—rather than an indirect frequentist one that can end up being misinterpreted and (d) encourage the reverse logic that asks what parameter values might have produced the data we see, rather than what data values will be produced by a parameter value.

ARTICLE HISTORY
Received May 2024
Accepted December 2025

KEYWORDS
Barriers; Bayes.
Test-inversion;
Mis-interpretations; Teaching

1. Introduction

Despite teachers having had nearly a century to explain confidence intervals (CI's) for parameters, misinterpretations persist (Greenland et al. 2016). As an example, consider a 95% frequentist CI for the speed of light, c, based on an already-calculated point-estimate $\hat{c}$ and a margin of error (*m.e.*): end-users often misinterpret it by saying that "c falls between $\hat{c}-m.e.$ and $\hat{c}+m.e.$ 95 times out of a 100." Less often written about are possible reasons why these misinterpretations continue, so I will draw on my own observations.

In part, this misconception arises because both the parameter and the data are typically expressed on the *same scale* (here metres/second). Thus, if object A is probably at most x units distant from object B, then object B is probably at most x units distant from object A. This arithmetical symmetry tends to blur the logical distinction between a parameter and a point estimate, between the fixed and the random. Words matter: an acorn may randomly *fall* near the tree, but the tree is *located* near an acorn: while standing, it does not *fall* near an acorn.

A second issue is that our teaching typically begins with *symmetric* intervals of the form $\hat{\theta} \pm m.e.$ This makes it difficult to then introduce the asymmetric ones that respect parameter boundaries when the space of the parameter θ (such as a proportion) is bounded, and the data are sparse.

Then, there is the matter of *semantics* and the *small print*: we rely on the frequentist behavior of the interval estimator; but, so as not to be accused of over-interpreting it (or slipping into making probabilistic statements about the parameter itself), we use the word "confidence" (Morey et al. 2016). To stress the role of sampling variation, the term *repeated sampling* is widely used. Unfortunately, this cryptic expression can give the impression that—even though we have expended our budget—*we* could repeat, or have "repeated" (on a new sample) the very investigation *we* have conducted. To explain sampling variation in words, it might be better to emphasize that we only got one shot at it, and that the sample result we just observed is just one realization of the near-infinite number of *possible results* that *could have been produced* by the play of chance. End-users sometimes make meaningless statements such as that "*this* estimate *is* accurate to within 3 percentage points 19 times out of 20" (Greenland et al. 2016). To explain in numbers, it is important to stress that the 19/20 refers to how well the inferential procedure performs over many applications: as when applied by sample survey organizations that conduct many surveys, or investigators who compute many confidence intervals over their lifetime. This explanation is nicely captured by the use of the plural: *samples of this size* produce estimates that *are* this accurate this often.

A further didactic difficulty, especially if CI's are introduced before hypothesis tests, is that several other new concepts are required in order to compute and understand the interval. These include the concept of a sampling distribution, its expectation and standard deviation (SD), what factors influence the shape of the distribution, and the adjustment one must make to the (t-based) number of standard errors to be used in the margin of error if components of it have to be estimated from the data in the sample. Sadly, when we have subsequently asked students to

CONTACT James A. Hanley *james.hanley@mcgill.ca* Department of Epidemiology, Biostatistics, and Occupational Health, McGill University, Suite 1200, 2001 McGill College Avenue, Montréal H3A 1G1, Québec, Canada.
Supplementary materials for this article are available online. Please go to *www.tandfonline.com/r/TAS*.
© 2026 American Statistical Association

recollect the essence of hypothesis tests and confidence intervals, they tend to conflate them with the context in which were first practiced, namely Gaussian and t-distributions. Our teaching of these concepts has not sufficiently divorced the inference concepts from the technical and mechanical aspects.

Lastly, even if the examples used involve data collected from students themselves (such as how many are left-handed, or how large their families are, or what fraction of the thumb-tacks they tossed fell point up) the inference is to some *larger universe* (of people, or families, or thumb-tack-tosses). And so, students may see the sample merely as a way to learn about a larger *external* entity or phenomenon that may not be of direct interest to them.

This note is a nonpartisan and multi-denominational appeal to teachers to address some of the misconceptions that students pick up from our teaching of confidence (and other-named) intervals for parameters. In support of our appeal, we describe an example of an integer-valued statistical parameter (estimand, quantity of interest) that (a) pertains to just the students and teacher in their own classroom (b) involves sensitive personal information that they may be uncomfortable sharing (c) can be estimated indirectly. The methods involve varying amounts of small print. Most importantly, the technicalities involved in computing the sampling distributions involved are kept out of the way—or already computed in a `Shiny` app—so that students can just focus on the logic and concepts.

We begin with how this specific inferential challenge arose: we describe the context in which the unknown values of that integer-valued statistical parameter became an over-riding personal concern for both teachers and students. In Section 3 we show how a combination of ingenuity and technology produces a statistic whose expectation parameter is linked—but not so obviously—to the parameter of concern. In Section 4 we show how parameter intervals for the latter can be computed indirectly by three approaches (a) Bayesian (b) the (prevailing) Method of Moments and (c) by inverting hypothesis tests. We end, in Section 5, by suggesting that (a) and (c) may create fewer misinterpretations than the prevailing one.

2. A Statistical Parameter of Personal Concern

In August 2021, the author and his colleagues were preparing their in-person graduate courses for classes ranging in size from 12 to 80 graduate students. All were fully vaccinated against COVID-19 and keen to know how many of their students were. However, the Instructor FAQs on our university's COVID-19 webpage included this edict:

> **Can I ask students in my class if they have been vaccinated?**
>
> No. Vaccination status is a confidential medical matter. In the Province of Québec, except in very limited circumstances that do not apply to instructors, it is illegal for an individual to ask students to disclose their vaccination status. This situation applies to professors and students. Please be reminded that vaccination is not mandatory in the Province of Québec.

Anonymous online polls are, possibly, one legal way to find out, but our university told us that "Surveys which do not guarantee the anonymity for all purposes are comparable to asking." In order to add a second layer of confidentiality for students and to keep the instructors further inside their legal boundaries, we considered an in-class online poll that uses a variant of the Randomized Response design (Genest, Hanley, and Bhatnagar 2024).

Randomized response designs were developed a half-century ago to allow participants in face-to-face interviews to answer sensitive questions on personal behaviors or history, while maintaining confidentiality. Survey topics have included drug use, tax evasion, abortions, and doping in sport.

Even if every student co-operates and truthfully answers the question(s) put to them, this design, by necessity, cannot provide an exact answer. However, its unusualness raises a surprisingly large number of teaching points, some of which are described in Genest, Hanley, and Bhatnagar (2024). But, in the context of parameter estimation, it also provides an illustration where it is possible to keep many of the (distracting) statistical technicalities out of the way and have the teacher and students focus on the essential concepts.

3. How This Parameter Modulates the Data Distributions in a Randomized Response Design

In our context, it is easy to specify the population and the estimand:

- *Population*: the n students attending my in-person class
- *Estimand*: how many of these students have been vaccinated. We denote this unknown number by the Greek letter θ, to emphasize that it is the *parameter* of interest—or, as one influential writer calls it, the *quantity* of interest (QOI).

The data are collected via k (≥ 1) rounds of an adaptation of Warner (1965). In each round, each student, independently of each other student and of each other round, uses a private randomization device to randomly determine whether (with probability π) (s)he is to answer the question of interest, or (with probability $1-\pi$) to answer the opposite (negation of the) question. Thus, if $\pi = 0.75$,

$$\begin{cases} \text{with probability 0.75, (s)he answers the Q:} \\ \quad \textit{Are you } \underline{\textit{V}}\textit{accinated? Yes/No} \\ \text{with probability 0.25, (s)he answers the Q:} \\ \quad \textit{Are you } \underline{\textit{Un}}\textit{-vaccinated? Yes/No} \end{cases}$$

and reports the answer in the anonymous in-class online poll.

The (sufficient) *statistic* comprises one number, the total number of "Yes" responses reported in the k rounds of the anonymous online poll. We will refer to this random variable, with support on the integers from 0 to nk inclusive, as Y, and the realization as y.

Prelude to the various approaches to inference regarding θ

Central to this task are the $(n+1)$ possible distributions of Y. The `Shiny` app allows one to interactively generate these for any combination of n, k, and π. For the purposes of this article, we chose $n = 10$, $k = (1, 4)$ and $\pi = 0.75$. Thus, the 11 alternately colored rows of Figure 1 show their locations and shapes if we carried out $k = 1$ (top) or $k = 4$ (bottom) rounds in a class of $n = 10$ students. The actual expected frequencies

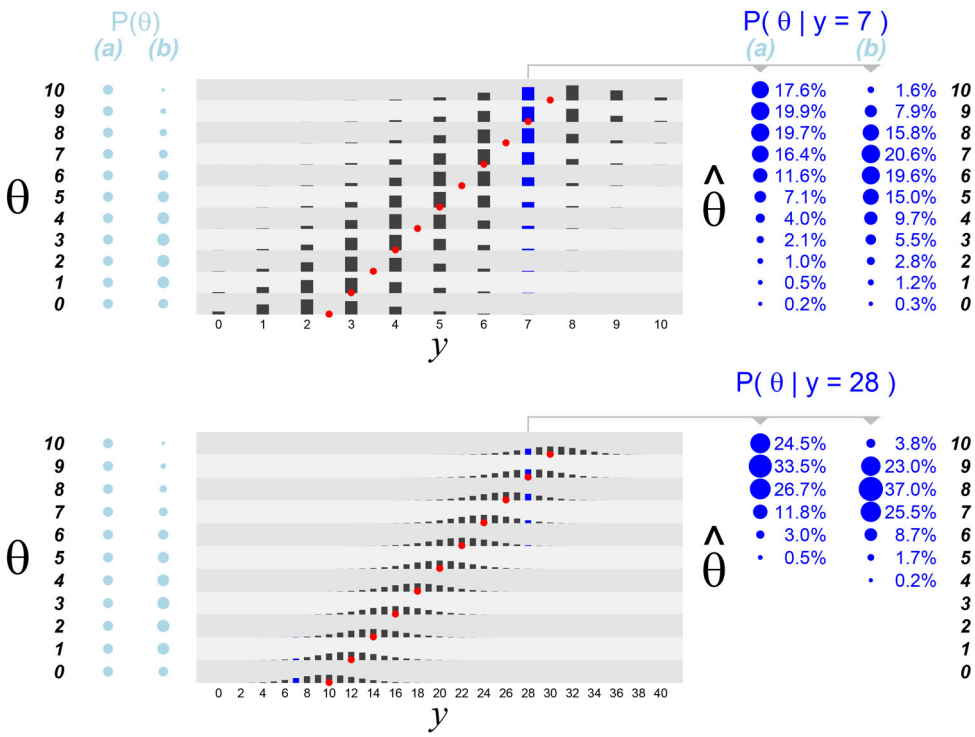

Figure 1. *Leftmost and rightmost, in black*: The $(n+1)$ possible values of the parameter θ, the unknown number of vaccinated students. *Inset, rowwise*: the $(n+1)$ possible distributions of Y, the total number of "Yes" responses if we carried out $k = 1$ (top) or $k = 4$ (bottom) rounds of the Warner version of the randomized response design in a class of $n = 10$ students, with π set to 0.75. *Left, in lighter blue*: The uniform and nonuniform solid circle sizes shown in columns (a) and (b), respectively, represent the uniform and nonuniform pre-data probabilities that will be used to compute the post-data probabilities shown at the right of the panel. [The expected value of $Y|\theta$ is a function of θ and (for use in Section 4.2) these expected values are shown as red dots; the value of θ where this expectation matches the observed value (y) is referred to as the Method-of-Moments estimate of θ.] *Column of rectangles, shown in blue*: The sizes of these rectangles in the $y = 7$ (20) column represent the *Likelihood* function $L(\theta|y = 7(28))$. *Right, circles shown in darker blue*: The post-poll (posterior) estimates of θ, after we have observed $y = 7$ (top) or 28 (bottom) "Yes" responses; they correspond to the uniform and nonuniform pre-data probabilities shown in the columns in lighter blue on the left.

for the single-poll case are given in a table in the Appendix in the supplementary materials. The sets of expected frequencies were computed in R from very basic principles, but in the spirit of not being derailed by the technical details, they too are located there. The other material in Figure 1 will be explained in the next section.

From these distributions, we can proceed inferentially in a number of ways. The order in which we chose to present them also happens to be the order in which they arose historically. For illustration, we suppose that in a survey of the $n = 10$ class attendees, there were $y = 7$ Yes and 3 No responses. From these data, what should the instructor infer as to θ, that is, as to how many of the n had been vaccinated? We will call the post-survey estimate $\hat{\theta}$.

4. Three Possible Statements About the Parameter

4.1. The Most Direct, and Least Convoluted, Approach (Bayesian)

We begin with an unrealistic scenario: imagine that the instructor had been "living under a rock" and was unaware of the progress or success of the vaccination campaign. Although there are various definitions of "ignorance," we will take it to mean that, *prior to the survey*, (s)he believed (or just assumed) that each of the $\{0, \ldots, n\}$ integer values of the parameter θ was equally likely to be correct. These 11 uniform probabilities are shown in the light blue (a) column in Figure 1. The chance that any particular *post-survey* $\hat{\theta}$ is correct can be calculated directly from first principles, that is, from the rules for conditional probabilities, commonly known as Bayes' theorem. Because the prior probabilities are uniform, the 11 posterior probabilities are directly proportional to the areas of the rectangles in the *Likelihood* column. These (scaled so they sum to 100) chances of being right are shown in blue in the (a) column in the rightmost end of Figure 1. Thus, from the 7/10 Yes responses the instructor has an 11.6% + 16.4% + 19.7% + 19.9% + 17.6% = 85% chance that (s)he is right if (s)he states that a *majority* ($\geq$ 6) of the 10 students are vaccinated. With more rounds of data, (s)he has an even higher chance that this statement about a majority is correct: for example, if after $k = 4$ rounds, there were a total of $y = 28/40$ Yes responses, the chance that a majority have indeed been vaccinated is approximately 99.5%; indeed 85% of the probability mass is concentrated on the integers 7, 8, 9, and 10.

In practice, the instructor would have been paying close attention to the vaccination program. Suppose her(his) pre-survey beliefs or assumptions regarding θ were described by the somewhat more pessimistic distribution shown in the lighter blue column (b) at the far left of Figure 1. The corresponding post-survey probabilities are shown in the darker blue (b) column in the rightmost end of Figure 1. *Computationally*, they can be seen as a re-weighting of the likelihood values with the pre-poll probabilities as weights; *philosophically*, they can be seen as a poll-based updating of the pre-poll probabilities.

A major part of the appeal of the Bayesian approach is that, with the prior set down, everything else is automatic. For example, one can make direct probability statements about the parameter θ using sentences in which (grammatically) the parameter is the *subject* of the sentence: we can speak or write of the probability that θ "is" (lies, or is located) in some numerical range. [In the prevailing (frequentist) approach, if we are to be linguistically and statistically correct, we should (but often fail to) speak of how many of the numerical ranges produced by the *procedure* are correct.] As a consequence, and as is illustrated by whether the majority of the class have been vaccinated, we can use the standard rules of probability to also directly address the probabilities associated with individual θ values or with any union of them. This is not possible in the frequentist approach.

4.2. The Prevailing Approach

Today, the most common approach in elementary statistics courses is to begin by using the realization of some statistic as a "*point estimate*," called $\hat{\theta}$, of θ; one then computes a "*standard error*" by using either the theoretical standard deviation of the (assumed to be Gaussian) sampling distribution of this statistic, or, if need be, a plug-in estimate of it. Finally, one computes a multiple of this *s.e.*, and designates it as the *margin of error* (*m.e.*). The resulting "point estimate $\pm$ *m.e.*" is called a *confidence interval*. The confidence level comes from the number of *s.e.*'s used in the *m.e.* For example, a Gaussian $z = 1.96$, or, if need be, in the case of a sample mean, the corresponding percentile of the "Student' t distribution, provides a 95% 'confidence level."

Hald (p. 20) tells us that "point estimate $\pm$ *m.e.*" intervals date back at least as far as 1776, when Lagrange gave what "seems to be the first example of a non-Bayesian probability statement about a parameter." Lagrange's concern was with a proportion parameter, where the obvious point estimate is the proportion of "Yes" responses in a sample that is large enough for the Central Limit Theorem to apply to the binomial sampling distribution. In our survey method, a point estimate is less obvious. In such cases, an easy-to-explain way to estimate the parameter is by the Method of Moments, which equates the empirical moment(s) to the theoretical moment(s). In our case, it involves writing the expected value $E[Y]$ as a function of θ, namely $(3/4)\theta + (1/4)(10-\theta)$; equating it with the observed value of the statistic ($y = 7$), and solving for θ, to obtain

$$\hat{\theta} = \frac{1}{(2 \times 3/4) - 1} \times (7 - (1/4)10) = 2 \times (7 - 2.5) = 9.$$

(One can use Figure 1 to find this, by locating the θ value (row) in which the $E[y|\theta]$ function matches the observed value $y = 7$.) As we show in the Appendix in the supplementary materials, the standard deviation of the sampling distribution of Y is $2 \times \sqrt{10 \times (1/4) \times (3/4)} = 2.74$, and so the 95% "point-estimate $\pm$ ME" interval for θ is $9 \pm 1.96 \times 2.74$, that is, 9 ± 5.5, or 3.5–14.5.

In this example, the "point-estimate $\pm$ *m.e.*" limits extend beyond the (0–10) range of the parameter; even with four rounds of the poll, the upper limit is 11.7. As can be seen in explorations using the `Shiny` app, this anomaly is common in small $n \times k$ scenarios. Thus, the method only provides an *approximate* confidence interval.

This difficulty is not limited to the peculiar (integral) nature of the parameter. It also applies to confidence intervals for more mainstream parameters, such as say the proportion, θ, of the Earth's surface that is covered by water, the estimation of which by a series of "random landing points" is an engaging classroom exercise (Gelman and Nolan 2017). The point estimate $\hat{\theta}$ is usually taken to be the observed sample proportion, and Andersson (2022) tells us that the "Wald" confidence interval, $\hat{\theta} \pm \hat{\theta} \pm z\sqrt{\hat{\theta}(1-\hat{\theta})/n}$ "can be traced as far back as Laplace (1812) and is surely the most well known (and still most used?) confidence interval for a binomial." Thus, if, at 5 randomly selected locations on the Earth's surface, we observed 4 covered by water, then the "point-estimate $\pm$ *m.e.*" 95% interval is 0.45 to 1.15. Even if the result were less extreme, say 3/5, or 2/5, the 95% CI would have extended past the boundaries for a proportion: 3/5 yields a Wald 95% CI of 0.17–1.03, and 2/5 yields one of -0.03 to 0.83. Indeed, *all six* possible results yield embarrassing intervals. One way out of these embarrassments is to compute a symmetric interval on an open-ended scale, such as the logit(p) scale, and then convert it back to the p scale. Another, where one shifts the symmetric interval in from the boundary by pretending it arose from an "observed" proportion of $(y + 2)/(n + 4)$, is actually an application of the reverse-engineering method we describe in this next subsection.

Quite aside from these (typically small-sample) difficulties, there is the wider issue of how easily and frequently the "point estimate $\pm$ margin of error" interval gets to be mis-interpreted. In most courses, the teaching of confidence intervals is preceded by an introduction to the Normal (Gaussian) distribution. Typically, in these preliminaries, one learns that approximately 68% (95%) of its values are within 1 (2) standard deviation(s) of the population mean μ. In the accompanying exercises, μ is considered *known* and the emphasis is on where a sample statistic is located in relation to this known parameter value: μ comes first, and the standard deviation of the sampling distribution is used to determine where a future $\bar{y}$ is (un)likely to "fall."

To move up to confidence intervals, textbooks typically start by describing these variations of $\bar{y}$ around μ, using $\bar{y}$ as the *subject* of the sentence, and μ as the static object. The sentence is then algebraicly re-arranged so that the calculated limits, $\bar{y} - s.e.$ and $\bar{y} + s.e.$ become the subject of the sentence, with μ remaining the object. However, as in our examples of the speed of light, and the tree and the acorn, end-users tend to mistakenly switch the subject and the object, and speak of the parameter as if it were dynamic, that is, "*falling*" within some distance of the point estimate.

In the next sub-section, we describe an alternative perspective. We do not begin with the known θ (or μ) that is used in preliminary chapters, but with having already observed a specific $\hat{\theta}$. The (reverse-engineering) task is to gauge how far below or above $\hat{\theta}$ the (still static) parameter θ *lies*. There are two scenarios: (i) the point-estimate may be an *under*estimate, having "landed" below θ. If this were the case, it would place θ on the upper side of $\hat{\theta}$, at a location we will call θ_{upper}. (ii) It may be an overestimate, having landed above it; this scenario would place θ on the lower side of $\hat{\theta}$, at a location we will call θ_{lower}. When, as in our application, the parameter θ is a proportion, the sampling distribution in scenario (ii), that is, the one where

$\theta = \theta_{lower}$, does not have the same-shape or spread as the one in scenario (i), that is, the one where $\theta = \theta_{upper}$: the distribution of over-estimates is not a mirror-image of the distribution of under-estimates. Typical teaching of confidence intervals begins with the mean parameter (μ), so that the distributions of over- and under-estimates are mirror-images of each other no matter the value of μ, and thus no matter the adopted values μ_{upper} and μ_{lower}. Starting with a more general approach would force the end-user to more carefully distinguish what is static but unknown—the parameter θ—from what is dynamic/variable—the statistic $\hat{\theta}$. Unlike in the case of the two above-mentioned objects A and B, the two are not logically interchangeable, even if they are arithmetically linked.

4.3. A Reverse-Engineering Approach & the Requisite Small Print

Wilson (1927) was one of the first to point out the "elliptical" nature of what he termed the "usual" inference, where the common statement of "probable inference" would be (and often today, still is!) to say that:

> The probability that θ lies outside its limits, $\hat{\theta} \pm ME$, is less than or equal to α.

whereas

> Really the chance that θ lies outside a specified range is either 0 or 1; for θ actually lies within that range or does not. It is the observed value y which has a greater or less chance of lying within a certain interval of its expectation. If the observer has had the hard luck to have observed a relatively rare event and to have based his inference thereon, he may be fairly wide of the mark.

To Wilson, an "improved" inference method could be achieved by

> reason[ing] as follows: There is some true value θ. The probability that an observation as extreme as y will occur [i.e., where y lies outside the limits implied by θ_{lower} and θ_{upper}] is less than or equal to α. This form of statement throws the emphasis upon the fallibility of a particular observation in respect to being typical of a general situation.

Today, this reverse engineering approach to producing an equal-tailed confidence interval is the basis for what is usually referred to as "test inversion." Whether one uses Gaussian approximations (as Wilson did) or the exact probabilities shown in Figure 1, it avoids the false dichotomy (and having to switch logic) between "small"- and "large"-samples, and provides intrinsic and non-symmetric θ limits when appropriate. We will take $\alpha = 0.05$, so that, with respect to these lower and upper θ limits, the observed y would be considered an over- or an under-estimate, respectively, lying as close to (but without surpassing the 2.5th or falling short of the 97.5th) the respective 97.5%-ile and 2.5%-iles as the discreteness of the y distributions allows.

The (**post**-poll) construction of the compatible parameter range is shown in Figure 2, a screenshot from the `Shiny` app.

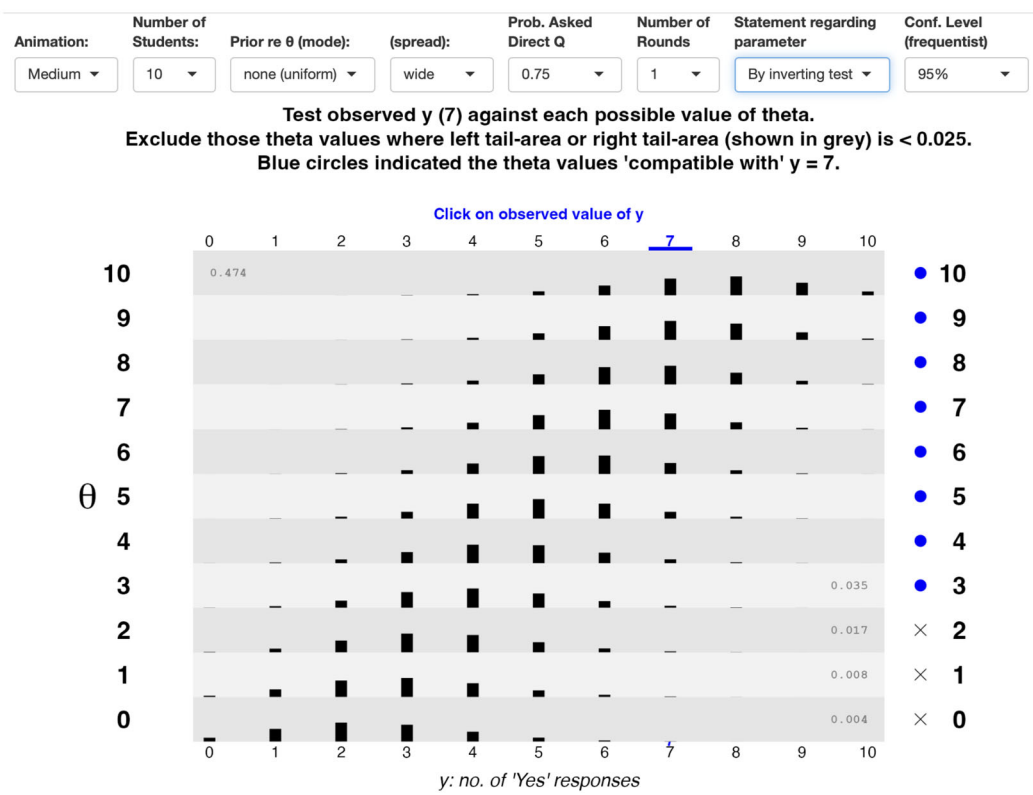

Figure 2. Screenshot from the `Shiny` app, illustrating the *inverted-test* approach to constructing the integer set for the parameter θ in the case of 7/10 Yes responses. One "tests" the observed y against increasing values of θ until, at $\theta_{lower} = 3$, the right-tail area first exceeds $\alpha/2$ (0.025). Likewise, to find θ_{upper} one proceeds through decreasing values of θ until (at $\theta_{upper} = 10$ here) the left-tail area first exceeds $\alpha/2$. The tail areas used to include/exclude each θ value are shown in grey. The included and excluded θ values are shown as blue circles and grey X's, respectively. The θ values 3–10 inclusive are deemed "compatible" with observing $y = 7$. If we observe 28/40 in $k = 4$ rounds, the range narrows to the integers 7–10 inclusive (distributions not shown, but available via the `Shiny` app).

Each possible θ value in turn is taken to be the "null" value. We exclude those null parameter values which, when tested against the observed y, produce a left or right tail area less than $\alpha/2$, and maintain those where both tail areas exceed it. (Just as with Clopper-Pearson intervals, the distribution in each row is discrete, and so coverage will be somewhat greater than the nominal level.)

Unlike with approach (a), this form of inference does *not* allow one to make a probabilistic statement as to whether (say) a majority (≥ 6) of the 10 students have been vaccinated. Nor is it possible to attach a separate probability to each element of the parameter set. Instead we are forced to regard the entire set as one and to quote the small print as to how, prospectively, the *procedure* will perform:

> In polls of classes of $n = 10$, the computed parameter set is expected to contain the (unknown) number vaccinated in the class in at least 95% of such polls.

This statement concerning the expected performance of at least 95% can be used to explain that the "confidence" is in the applications of the procedure, and that it applies to any value of n and their associated values of θ.

5. Ending Remarks

Despite devoting considerable class time to explaining how to and how not to interpret a traditional (frequentist) confidence interval, by the end of the semester many students seem to revert to mis-interpretations they came in with, or hear all around them, or can easily find on the internet. Among the worst examples from the author's experience [see, supplementary file] were (i) in the case of an observed percentage derived from an opinion poll of $n = 1000$: "*this* estimate is accurate to within 3 percentage points 19 times out of 20" and (ii) in the case of an 95% interval based of several replicate measurements of the speed of light: "I can be 95% sure that the speed of light falls between these calculated limits."

The aim here was to promote probability statements that respect the range of the parameter of interest, that employ the same logic for all sample sizes, that don't make the point estimate the center of attention, and that are less likely to be misinterpreted. To this end, the example was deliberately chosen so that the point estimate was not an obvious function of the observed data (in the example used by Clopper and Pearson, the point estimate of the population proportion is the sample proportion). This way, students are less likely to interchange the roles of the statistic and the parameter: one cannot treat them as though they are two objects of the same type, where the distances from one to each other are algebraically symmetric. We also tried be parsimonious, so that each method fits all sample sizes.

Rather than have students first master the considerable mathematical-statistics theory that governs sampling distributions, we prefer to present them with already-computed distributions, so that—with say the aid of an interactive tool like a `Shiny` app—they can just focus on the logic and concepts involved in making probability statements concerning parameters.

All too often, examples involve a parameter that describes a universe that is mostly or entirely outside the classroom. The statistical tools, instruments and techniques become more relevant when the parameter is personal—such as one's own mean or median reaction time, or how much one's activity levels changed during COVID, or is limited to some aggregate value concerning just those in one's class.

A number of considerations argue for presenting the approaches in the order in which they developed historically. The fact that the Bayesian approach was developed first (Bayes and Price 1764), is not a reason per se to give it priority. More compelling is that it unequivocally [with no small print or hiding behind the word "confidence"] and directly addresses the parameter of interest: it puts it as the subject in the sentence in a natural way. It places less emphasis on one "best" point-estimate, and instead allows for statements about individual, or subsets of the, possible θ values; and it legitimately places less credence on low-probability values.

Moreover, Bayes' theorem will have already been introduced when addressing unconditional and conditional probabilities. One of the commonest applications of it is the post-test interpretations of the results of medical tests, where the parameter of interest has just a 2-point distribution, for example, "not infected" ($\theta = 0$) or "infected" ($\theta = 1$). Examples such as ours make for a smoother transition to parameters that can take on more possible values. Many exercises surrounding medical tests simply supply the students with a prevalence value and ask them to use it as a prior; our example can lead to discussion about each individual's pre-poll sense of the class θ and to emphasize that student to student variation in their pre-poll—and thus post-poll—estimates of θ is natural.

If a teacher prefers to present just a frequentist approach, we would urge that it be the reverse-engineering (Wilson-inspired) test-inversion we employed in 4c. This will save having to have separate versions for small- and large-sample contexts, and reduce the chance of fuzzy or downright inappropriate interpretations. 2-D diagrams where the parameter values are on one axis, and the possible values of the statistic continue to be used in textbooks. However, unlike in the orientational example in Figure 9.2.1 in Casella and Berger (2002), the theoretical moments in our example do not fall on the 45 degree line, and the vertical and horizontal ranges of parameter and statistic differ in the $k > 1$ case. These special features of this example also emphasize that the parameter and the statistic can neither be arithmetically nor logically interchanged.

Over the years, we have increasingly steered away from the use of the concept of repeated sampling in pursuit of the same θ, and instead, as Wasserman (2004) does, explained that the coverage or misses should be averaged over all applications of such intervals, where the θ and the context are different each time. For example, in our vaccination example, θ will differ from class to class.

We typically begin our interpretation by saying that what we will observe (or have observed) will be (is) either an over-estimate (i.e., the truth is lower) or an under-estimate (i.e., the truth is higher). The reverse-engineering approach may seem more tedious, but it is more parsimonious. It yields non-symmetric parameter intervals when they are needed; and even when they are not, or when the sample size is so large that they

are effectively symmetric, it reminds the user that it is the *point estimate* that has fallen above or below the (hidden) true value of parameter, rather than the *converse*.

Even though all introductory textbooks introduce Bayes' theorem, most shy away from teaching credible intervals side by side with the traditional, but more legalese-demanding, "confidence" intervals. We hope that our example is seen to be amenable to a bi-partisan approach, and that teachers will supply their own examples of closer-to-home parameters—or, as Gelman calls them, quantities of interest. Rather than adopting a denominational stance, we appeal to teachers to adopt the wider use of both the Bayesian and inverted-test approaches to statements about these parameters. And we encourage them to share their suggestions on how to promote frequentist language that is accurate and useful, but not overly legalistic.

Supplementary Materials

The online supplementary material consists of (i) EndNotes (ii) details on computing the distributions of Y given theta (iii) a link to, and screenshots from, the Shiny app, and (iv) a statement that the R code for Figure 1 and for the Shiny app, along with some extra teaching material, can be accessed at this website *https://osf.io/cp738/overview?view_only=5b3e73821e7747ab9158ebc8b23e07b7*.

Acknowledgments

I thank (i) the 12 students in our graduate program in biostatistics who, in my in-person class in Fall 2021, let me pilot the variation of the randomized response design. As I explain above, our university instructed teachers not to ask students whether they had been vaccinated against COVID-19 (ii) Christian Genest for suggesting the simpler Method of Moments estimator. Although I do not cover it here, I include it in the Shiny app the Maximum Likelihood approach that I had suggested initially; as he pointed out, it is not a suitable topic for an introductory statistics course. (iii) Sahir Bhatnagar. We had many conversations about the test-inversion approach that he adopted in his in-person teaching confidence intervals for proportions to the 80 graduate students in epidemiology and public health that Fall. He also suggested wider uses of the randomized response design. (iv) the very patient and very helpful editors and referees for the large amount of work they devoted to this piece. I also acknowledge the considerable help from ChatGPT in coding the `Shiny` app.

Disclosure Statement

I have no competing interests to declare.

Funding

None.

ORCID

James A. Hanley https://orcid.org/0000-0003-3486-5291

References

Andersson, P. G. (2022), "Approximate Confidence Intervals for a Binomial p – Once Again," *Statistical Science*, 37, 598–606. DOI:10.1214/21-STS837 [4]

Bayes, T., and Price, R. (1764), "An Essay towards Solving a Problem in the Doctrine of Chance," (Foreward and Appendix added by Richard Price), *Philosophical Transactions of the Royal Society of London*, 53, 370–418 [6]

Casella, G., and Berger, R. L. (2002), "Section 9.2.1 (Inverting a Test Statistic)," in *Statistical Inference* (2nd ed.), Pacific Grove: Duxbury Press. [6]

Clopper, C., and Pearson, E. S. (1934), "The Use of Confidence or Fiducial Limits Illustrated in the Case of the Binomial," *Biometrika*, 26, 404–413.

Gelman, A., and Nolan, D. (2017), *Teaching Statistics: A Bag of Tricks* (2nd ed.), Oxford: Oxford University Press. [4]

Genest, C., Hanley, J. A., and Bhatnagar, S. R. (2024), "Investigating Sensitive Issues in Class Through Randomized Response Polling," *Journal of Statistics and Data Science Education*, 32, 457–463. DOI:10.1080/26939169.2024.2302179 [2]

Greenland, S., Senn, S. J., Rothman, K.J., Carlin, J. B., Poole, C., Goodman, S. N., Altman, D. J. (2016), "Statistical Tests, P values, Confidence Intervals, and Power: A Guide to Misinterpretations," *European Journal of Epidemiology*, 31, 337–350. DOI:10.1007/s10654-016-0149-3 [1]

Hald, A. (2004), "A History of Parametric Statistical Inference from Bernoulli to Fisher, 1713 to 1935," Department of Applied Mathematics and Statistics, University of Copenhagen.

Morey, R. D., Hoekstra, R., Rouder, J. N., Lee, M. D., Wagenmakers, E.-J. (2016), "The Fallacy of Placing Confidence in Confidence Intervals," *Psychonomic Bulletin & Review*, 23, 103–123. DOI:10.3758/s13423-015-0947-8 [1]

Warner, S. L. (1965), "Randomised Response: A Survey Technique for Eliminating Evasive Answer Bias," *Journal of the American Statistical Association*, 60, 63–69. DOI:10.1080/01621459.1965.10480775 [2]

Wasserman, L. (2004), *All of Statistics: A Concise Course in Statistical Inference*, New York: Springer. [6]

Wilson, E. B.(1927), "Probable Inference, the Law of Succession, and Statistical Inference," *Journal of the American Statistical Association*, 158, 209–212. [5]